

MATERIALS SCIENCE

Noncoplanar multidentate contacts reinforce buried interface for perovskite/silicon tandem solar cells

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The buried interface between the self-assembled monolayers (SAMs) and the perovskite plays a critical role in device performance and stability in inverted perovskite solar cells (PSCs). However, severe desorption and agglomeration of self-assembled molecules lead to interfacial losses and shorten device longevity. Herein, we design a molecule, 5,5',5''-(nitrilotri-4,1-phenylene)tris[2-thiophenecarboxylic acid] (TTA) with a noncoplanar molecular configuration and multiple functional groups. The noncoplanar molecular configuration could effectively suppress π - π stacking-induced aggregation and enable more regulated molecular packing. Its thiophene functional unit and tridentate carboxylic anchoring groups enhance interactions with substrates and the upper perovskites, assisted by the noncoplanar configuration, which is confirmed by density functional theory (DFT) calculations. When combined with the commonly used 4-(3,6-dimethyl-9H-carbazol-9-yl)butane-1-phosphonic acid (Me-4PACz), this strategy enables a more homogeneous self-assembled molecular film, improved perovskite crystallinity, and reduced trap density. Consequently, the wide-bandgap (WBG) PSC (~ 1.67 electron volts) demonstrates a champion power conversion efficiency (PCE) of 23.7%, with enhanced thermal-cycling and light-soaking stability. In addition, the perovskite/silicon tandem devices based on this strategy achieve a PCE of 33.3% (certified at 33.1%) and demonstrate real-world operational stability of no observed PCE loss after 30 days of outdoor operation.

INTRODUCTION

Wide-bandgap (WBG) perovskites with a bandgap of about 1.65 to 1.70 eV play a pivotal role in high-efficiency perovskite/silicon tandem devices, enabling higher open-circuit voltages and favorable current matching with silicon bottom cells (1–5). The emergence of self-assembled monolayers (SAMs) reduces interfacial losses in inverted devices, thereby further improving device power conversion efficiency (PCE) (6–8). However, certain challenges still exist in SAMs. For instance, the widely used carbazole-based SAM molecule 4-(3,6-dimethyl-9H-carbazol-9-yl)butane-1-phosphonic acid (Me-4PACz) exhibits excellent hole-extraction capability but suffers from self-aggregation when processed from alcoholic solvents (9, 10). This aggregation results in uneven SAM coverage and localized island formation with pinholes, which weakens the interface binding to both substrates beneath and the perovskite layer, increases interface defect density, and impairs charge transport (11–13).

Various reports highlighted the critical roles of coverage uniformity and strong interface binding in achieving both high performance and long-term stability (14, 15). Introducing a mercapto-functionalized scaffold effectively regulates buried-interface crystallization and defect density, yielding strongly bonded interfaces that reduce carrier losses and enhance device performance (16). Modifying the phosphonic

anchor or the molecular linker to tune interface energetics and binding strength is a typical strategy, which can improve level alignment and SAM/substrate coupling (17, 18). Besides, introducing steric hindrance side groups (e.g., bulky *tert*-butyl) to suppress intermolecular aggregation is also effective, forming more uniform layers and reducing voids at the hole transport layer (HTL)/substrate interface (8, 19, 20). These methods effectively alleviate one specific issue; however, delivering multiple benefits simultaneously with a single molecule, especially for WBG perovskites, is less discussed. There is still considerable room for improvement in the PCE and stability for WBG perovskite solar cells (PSCs) and perovskite/silicon tandems.

To address the above issues, we designed a molecule, 5,5',5''-(nitrilotri-4,1-phenylene)tris[2-thiophenecarboxylic acid] (abbreviated as TTA), as shown in Fig. 1A. The molecular design of TTA integrates the following features. First, TTA adopts a noncoplanar molecular configuration, which suppresses excessive π - π stacking between adjacent molecules and disrupts the aggregation behavior typically observed in coplanar conjugated SAM systems (21). Second, tridentate structures of carboxyl groups ($-\text{COOH}$), assisted by a noncoplanar structure, allow free rotation and enable a more favorable multisite binding probability to the nickel oxide (NiO_x) substrate and perovskite buried interface, further contributing to defect passivation and improving wetting properties (22). Third, incorporation of thiophene units further extends the delocalized π -electron system, regulating π - π interactions and interfacial binding energy (23). These complementary interactions thereby facilitate suppression of Me-4PACz self-aggregation, resulting in enhanced interface quality, reduced trap density, and improved perovskite film formation. When combined with Me-4PACz, this strategy further promotes high-quality perovskite film formation and enhanced device performance. The WBG PSC demonstrates a champion PCE of 23.7%, with enhanced thermal-cycling (TC) and light-soaking stability. In addition, the perovskite/silicon tandem devices achieve a PCE of 33.3%

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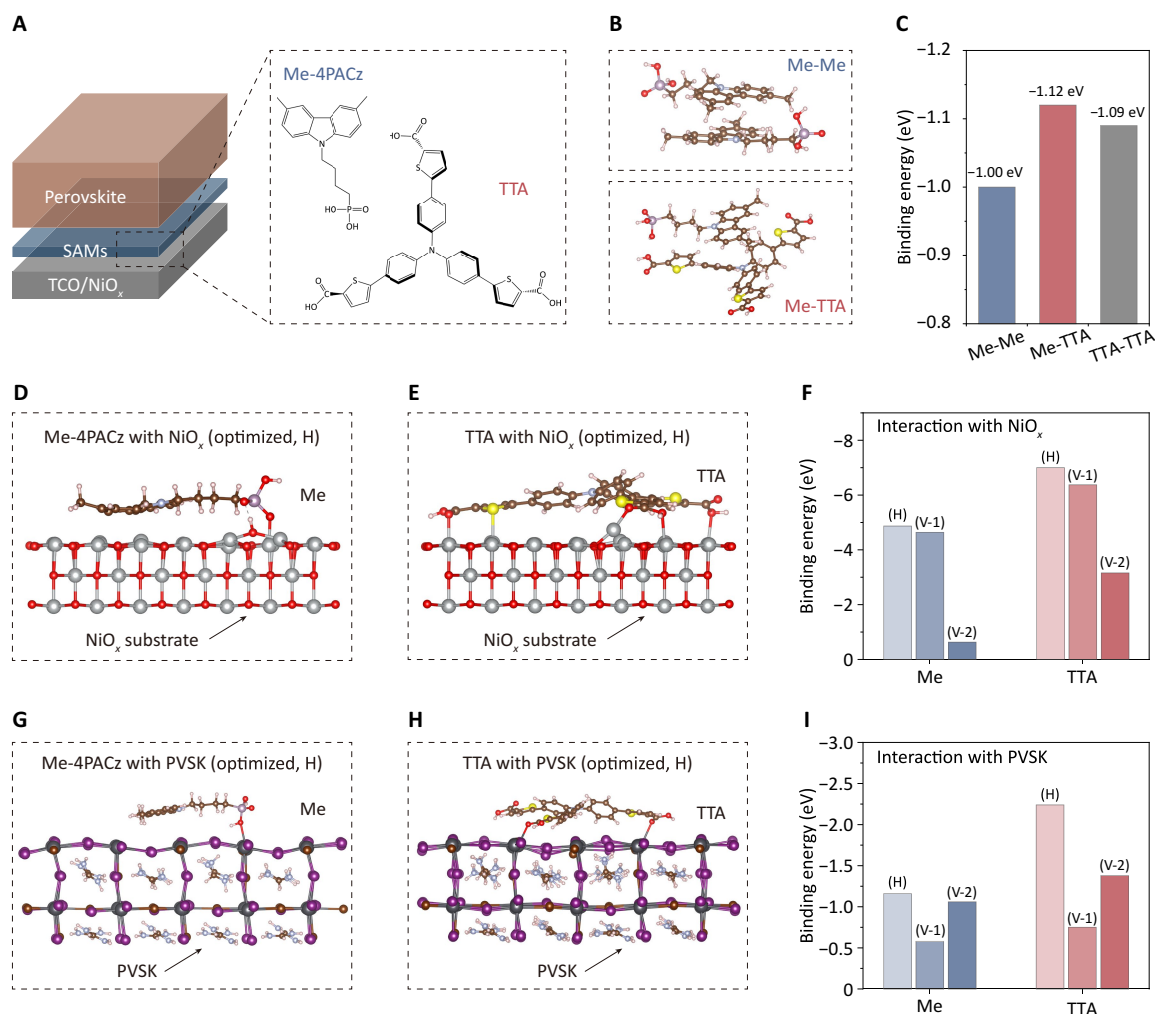


Fig. 1. Schematic illustration and calculated binding interactions of Me and TTA. (A) Device configuration illustration and chemical structures of Me and TTA. (B) Optimized configurations of Me-Me and Me-TTA pairs. (C) Calculated binding energies of Me-Me, Me-TTA, and TTA-TTA pairs with different configurations. (D to F) Optimized horizontal configurations and binding energies of Me and TTA molecules with the NiO_x substrate. (G to I) Optimized horizontal configurations and binding energies of Me and TTA molecules with the perovskite layer.

(certified at 33.1%) and show no PCE loss after 30 days of outdoor operation, demonstrating superior real-world stability. These findings indicate that rational interface engineering via hybrid SAMs can effectively optimize both efficiency and long-term durability in PSCs.

RESULTS

To illustrate the design principle of our strategy, we first used density functional theory (DFT) to prove interactions between different molecules and their aggregations. For convenience, Me-4PACz is abbreviated as Me thereafter. It is worth noting that, unlike previously reported coplanar aromatic modifiers or single-functional-group molecules, TTA has a noncoplanar molecular configuration with a thiophene unit and three carboxylic anchoring groups, as shown in fig. S1, which is expected to simultaneously regulate intermolecular packing and favor interfacial interactions. Figure 1B presents the optimized configurations of Me-Me and Me-TTA pairs

(the configuration of TTA-TTA is shown in fig. S2). In Fig. 1B, Me-4PACz prefers a parallel arrangement when self-aggregates, consistent with previous reports (17, 24). Such aggregation behavior is primarily driven by strong π - π stacking interactions between planar conjugated backbones. In contrast, the noncoplanar and twisted geometry of TTA can reduce effective π - π overlap, thereby disrupting the ordered stacking of Me-4PACz molecules when incorporated into the SAM. Figure 1C summarizes the possible intermolecular interactions in the hybrid SAM, including Me-Me, Me-TTA, and TTA-TTA. The calculated binding energies are -1.00 eV for Me-Me, -1.12 eV for Me-TTA, and -1.09 eV for TTA-TTA. The stronger Me-TTA interaction indicates that incorporating TTA molecules can effectively break the self-aggregation of Me-4PACz molecules and promote the formation of a uniform SAM layer.

We then used DFT to further investigate interfacial binding in the presence of TTA molecules. We examined three possible types of molecule configurations on the NiO_x substrates: one horizontal and two vertical orientations (denoted as H-1, V-1, and V-2), which are

displayed in Fig. 1 (D and E) and fig. S3A. The calculated binding energies with NiO_x for different configurations are summarized in Fig. 1F. For both molecules, the horizontal structure is preferred and has the highest binding energy: -4.87 eV for Me-4PACz and -6.98 eV for TTA, respectively. In addition, compared with Me-4PACz, the TTA molecule exhibits higher binding energy in all three cases, indicating a stronger interfacial adhesive force with the underlying substrate.

Similarly, Fig. 1 (G to I) illustrates the interactions between the SAM molecules and the perovskites, with Fig. 1 (G and H) and fig. S3B showing all the possible geometries. The calculated results for different configurations are shown in Fig. 1I, where Me-4PACz and TTA molecules also prefer a horizontal adsorption configuration, with the highest binding energies of -1.16 eV for Me-4PACz and -2.24 eV for TTA, respectively. Considering that the perovskite surface may also be terminated by organic cations, we further constructed a model with FA^+ -terminated surfaces to evaluate the interaction with SAM molecules. As shown in fig. S4, the calculated binding energy of TTA on the FA^+ -terminated surface is -4.47 eV, which is notably higher than that of Me-4PACz (-2.39 eV), indicating a stronger interaction between TTA and the organic surface components. This result suggests that, beyond the O-Pb^{2+} coordination considered above, TTA can also effectively interact with organic surface species, thereby further stabilizing the hybrid SAM/perovskite interface.

To rationalize the origin of the enhanced interfacial interaction, we further analyzed the electronic properties of the interface. The multifunctional nature of TTA plays a critical role, where the conjugated thiophene unit can participate in intermolecular π interactions with the aromatic framework of Me-4PACz, whereas the three carboxylic groups enable multidentate anchoring to the substrate, collectively stabilizing the molecular arrangement within the SAM. Charge density difference calculations (figs. S5 and S6) reveal a more pronounced charge redistribution at the interface in the presence of TTA, indicating stronger electronic coupling between the SAM and adjacent layers. Moreover, the integrated charge transfer along the z direction shows a larger amount in the TTA-containing system than in the pristine Me-4PACz case, further confirming the strengthened interfacial electronic interaction.

These results collectively indicate that the synergistic effect of the electron-rich thiophene unit and multidentate carboxylic groups enhances interfacial electronic coupling, which is consistent with the DFT binding energy analysis. According to the results, the TTA molecules exhibit stronger binding energy with both the NiO_x and perovskite layers than Me-4PACz, indicating enhanced interfacial coupling and potential to improve the overall device stability.

To further investigate the composition and interfacial properties of the SAM layers, a series of characterizations was performed. Here, the Control refers to samples with a pure Me-4PACz, whereas the Target refers to samples fabricated with a hybrid SAM composed of Me-4PACz and TTA in a 3:1 molar ratio. As shown in Fig. 2, the x-ray photoelectron spectroscopy (XPS) spectra of the characteristic sulfur (S) element (Fig. 2A) confirm the successful incorporation of TTA molecules in the Target SAMs with the structure of indium tin oxide (ITO)/ NiO_x /SAMs because sulfur is the only element present in TTA. It is worth noting that the anchoring strength of different functional groups may influence the formation of the hybrid SAM. DFT calculations show that a single $-\text{PO}(\text{OH})_2$ group in Me-4PACz (V-1, -4.64 eV) binds more strongly than a $-\text{COOH}$ group in TTA

(V-2, -3.16 eV) on the NiO_x surface (Fig. 1F). However, the overall interaction is dominated by collective coordination, where the three carboxylic groups in TTA enable multidentate binding and enhance the overall adsorption stability. To verify that the $\text{p}K_a$ (where K_a is the acid dissociation constant) differences of phosphonic acid and carboxylic acids do not lead to molecular displacement during codeposition, XPS analysis was performed (fig. S7). The P:S atomic ratio (nearly 1:1) agrees well with the precursor composition, confirming that TTA is stably incorporated into the hybrid SAM. Time-of-flight secondary ion mass spectrometry (TOF-SIMS) further confirms the presence of TTA at the interface without diffusing into the perovskite (Fig. 2B and fig. S8). The cross-sectional transmission electron microscopy (TEM) images shown in Fig. 2C reveal that the pure Me-4PACz layer exhibits nonuniform coverage. To exclude the influence of substrate heterogeneity, multiple batches of samples were examined, and TEM images were collected from different regions (figs. S9 and S10), which consistently show that the Control samples exhibit discontinuous coverage even on relatively smooth substrates. In contrast, the hybrid SAM layer shows a more homogeneous morphology with an apparent thickness of about 5 nm, suggesting a multilayer formation rather than a strict monolayer. The improved continuity suggests that TTA molecules suppress Me-4PACz aggregation and promote the formation of a more uniform interfacial layer.

To further confirm the dense and uniform coverage after TTA addition, we performed conductive atomic force microscopy (C-AFM) to map surface conductivity. As shown in Fig. 2 (D and E), the Control sample with the ITO/ NiO_x /Me-4PACz structure displays numerous bright regions corresponding to high-current leakage signals, indicative of uncovered areas in the SAM layer. In comparison, the Target sample shows fewer high-current regions, suggesting that the hybrid SAM forms a denser, more continuous interfacial layer (8, 25). To further probe the lateral chemical homogeneity of the SAM layers, Fourier transform infrared spectroscopy (FTIR) and atomic force microscopy-infrared (AFM-IR) mapping were performed. As shown in fig. S11, the characteristic vibrational signals of Me (1105 cm^{-1}) and TTA (1665 cm^{-1}) exhibit uniform spatial distributions in the hybrid SAM, indicating that TTA is homogeneously incorporated without forming localized aggregation domains. Compared with the target sample, the control shows a weaker and less uniform distribution of the Me signal, suggesting molecular aggregation and partial desorption. Consistently, AFM measurements (fig. S12) reveal a smoother surface morphology with reduced roughness after introducing TTA (from 0.51 to 0.34 nm), further supporting the formation of a more homogeneous and well-regulated interfacial layer. These results collectively demonstrate that the incorporation of TTA effectively suppresses the aggregation of Me-4PACz and improves the lateral uniformity of the hybrid SAM. Moreover, grazing incidence wide angle x-ray scattering (GIWAXS) measurements were performed to further investigate molecular packing and structural ordering. As shown in fig. S13, the diffraction patterns of the control and target samples are largely similar, with no obvious changes in peak position or intensity. This suggests that the incorporation of TTA does not notably alter the long-range ordering of the SAM or the overlying perovskite layer.

The desorption of SAMs under thermal conditions is one of the primary factors that deteriorate the device's longevity. To demonstrate enhanced binding properties under stress, Kelvin probe force microscopy (KPFM) was used to measure the surface potential of

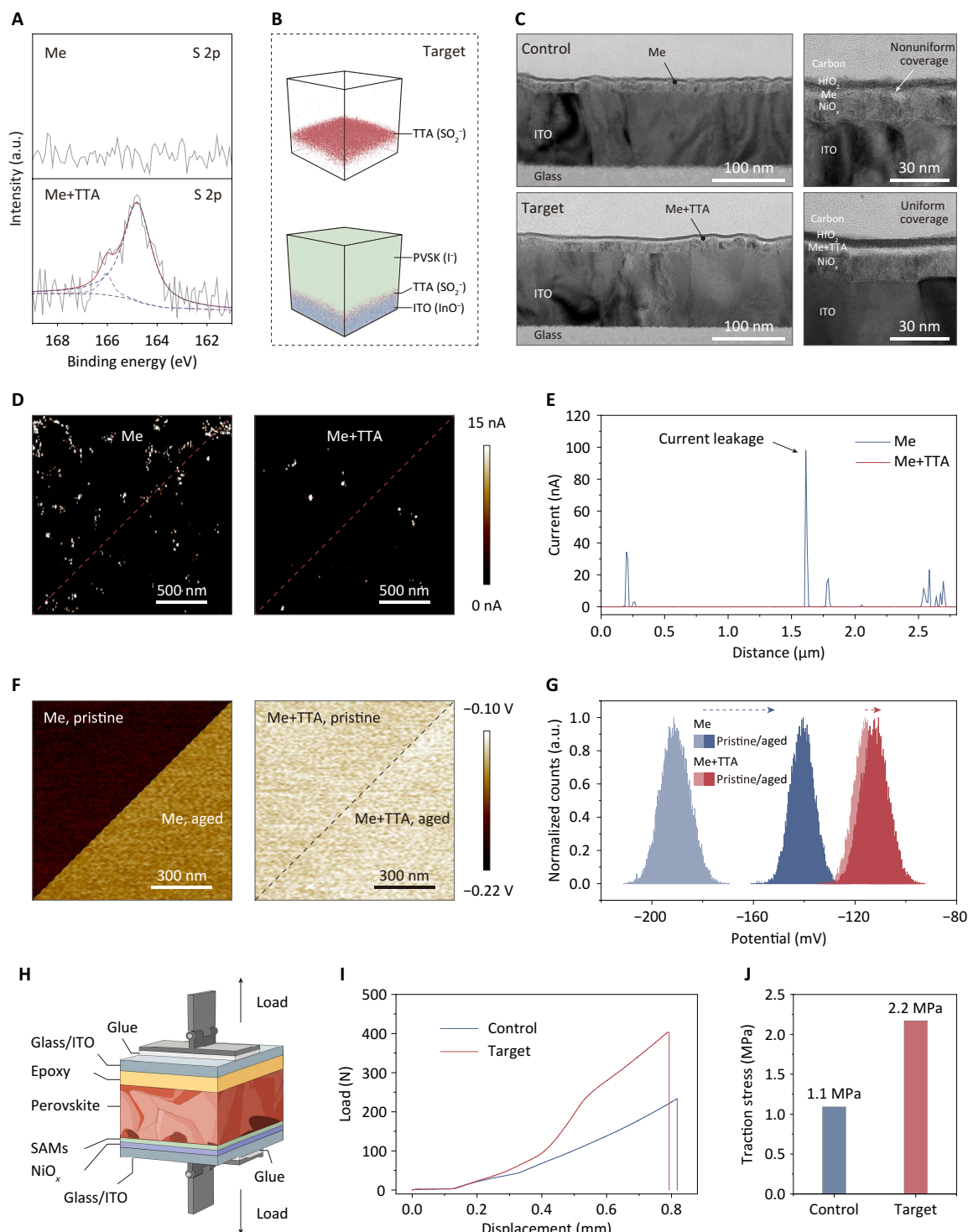


Fig. 2. Structural and interfacial characterizations of SAM layers. (A) XPS spectra of the S 2p region for control pure Me-4PACz and target hybrid SAMs. a.u., arbitrary units. (B) TOF-SIMS depth profiles of target SAM layers in the ITO/ NiO_x /SAM/perovskite stack. (C) Cross-sectional TEM images of the control pure Me-4PACz and target hybrid SAMs. (D and E) C-AFM current mapping of control pure Me-4PACz and target hybrid SAMs. (F and G) KPFM images of the control pure Me-4PACz and target hybrid SAMs before and after thermal aging at 110°C for 8 hours. (H to J) Schematic illustration and results of the interfacial tensile strength.

the SAM layers. The mappings of SAM layers on ITO/NiO_x substrates before and after aging at 110°C for 8 hours are shown in Fig. 2 (F and G). Given that some studies evaluate SAM thermal stability by applying long-duration thermal aging at around 100°C, we similarly subjected our samples to this treatment (26). After heating, the Control sample shows a much larger potential variation (from −192 to −141 mV). In contrast, the Target sample maintains a relatively stable surface potential, confirming its superior thermal durability. In addition, the enhanced binding strength between SAM molecules and the NiO_x substrate was further evaluated by solvent washing. Variations in XPS signal intensities before and after ethanol (EtOH) rinsing were compared in fig. S14. The N peak of the Target sample decreased less notably than that of the Control, suggesting stronger binding interactions between the hybrid SAM and NiO_x. Last, according to DFT simulations (Fig. 1, G to I), the binding also enhances with the perovskites after the addition of TTA. To elucidate interfacial interactions between SAMs and perovskite precursors, XPS was performed on thin PbI₂ layers deposited on different SAM substrates (fig. S15). Compared with PbI₂ on Me, the Pb 4f spectrum of PbI₂ deposited on the Me+TTA substrate shows a more pronounced binding energy shift, indicating stronger electronic coupling at the hybrid interface. The interfacial tensile strength is a typical demonstration of the interfacial reinforcement (Fig. 2H) (27, 28). As shown in Fig. 2 (I and J), the Target sample exhibits a markedly higher tensile strength of 2.2 MPa compared with 1.1 MPa for the Control, demonstrating that TTA incorporation substantially enhances the weak interfacial binding between the SAM and the perovskite layer (27, 28).

To further evaluate the influence of hybrid SAM on perovskite crystallization, both top and bottom perovskite surfaces were characterized by scanning electron microscopy (SEM) (Fig. 3, A to D). The top surfaces of the Control and Target samples appear similar, with more uniform, but relatively smaller, grains than in the Control (29). This behavior can be attributed to the improved wettability of the hybrid SAM, which increases the nucleation density at the perovskite/SAM interface, leading to reduced grain size (fig. S16). The bottom surfaces of the Control and Target perovskite films exhibit clearer, more uniform grains, suggesting that the hybrid SAM promotes more homogeneous crystallization. The improved crystallinity was further confirmed by x-ray diffraction (XRD) analysis (Fig. 3E), where the Target sample exhibited a stronger (100) diffraction peak (at about 14.2°) than the Control (30). We conclude that the crystallinity enhancement results from the improved wettability (fig. S16) and reduced colloidal particle size of the hybrid SAM. Dynamic light scattering (DLS) measurements (fig. S17) reveal single-modal size distributions for both samples at a low concentration, with the Target solution exhibiting a narrower, left-shifted peak (about 200 to 300 nm) relative to the Control (about 600 to 900 nm), indicating a reduced average particle size. The polydispersity index (PDI) decreases notably from 0.294 for the Control to 0.164 for the Target, confirming improved uniformity of the colloidal particles (31). It should also be noted that the observed sizes mainly arise from dynamic aggregates rather than individual molecules. Additional measurements at a higher concentration (8.00 mg/ml) (fig. S18 and table S1) reveal the presence of sub-10 nm species, indicating that the precursors primarily exist as small molecular species or clusters in solution. These results suggest that TTA modulates the aggregation behavior, leading to a more uniform precursor distribution.

Photoluminescence (PL) mapping measurements of the bottom surfaces (Fig. 3, F and G) of the Target perovskite film exhibit

enhanced intensity, suggesting improved interfacial homogeneity and reduced defect variation across the surface (30). In addition, PL mapping measurements of the top surface (fig. S19) reveal a more uniform emission distribution for the Me+TTA-treated films, suggesting improved interfacial homogeneity and reduced defect variation across the surface. To investigate carrier dynamics, time-resolved photoluminescence (TRPL) measurements were conducted (Fig. 3H and table S2). The results show that the Target film exhibits a pronounced prolongation of both the fast and slow decay components compared with the Control. The fast component τ_1 , commonly assigned to the nonradiative interfacial recombination, increases from 29.6 ns (Control) to 97.8 ns (Target). At the same time, the decrease of its amplitude A_1 indicates a strong suppression of trap-assisted recombination. The slow component τ_2 (assigned to the interfacial and bulk recombination) is prolonged from 1178.1 to 1944.0 ns, and the overall average lifetime τ_{avg} increases from 1170.1 to 1933.1 ns (32, 33). These changes demonstrate that the hybrid SAMs effectively passivate interfacial traps and improve bulk recombination properties (34). To further distinguish the contributions of carrier recombination and interfacial charge extraction to the observed PL dynamics, transient absorption (TA) spectroscopy was performed (fig. S20). The time-resolved TA spectra of the Target sample exhibit a gradual upward shift of the ground-state bleaching (GSB) minimum with increasing delay time, indicating a recovery of the band-edge state filling. This behavior suggests that charge carriers are extracted for an ultrashort period, unlike those in the control sample, which are hindered. Combined with the enhanced steady-state PL intensity, these results confirm that the prolonged carrier lifetime primarily originates from suppressed defect-assisted nonradiative recombination rather than interfacial transport limitations. To quantify the effect of the hybrid SAM on defect passivation, space-charge-limited current (SCLC) measurements were performed on hole-only devices (fig. S21). The trap-filled limit voltage (V_{TFL}) value decreases from 0.71 V for the Control device to 0.67 V for the Target device, corresponding to a reduction in defect density (N_{trap}) from 1.41×10^{16} to $1.31 \times 10^{16} \text{ cm}^{-3}$. Last, ultraviolet photoelectron spectroscopy (UPS) measurements (Fig. 3, I and J) show that the ionization energy difference (ΔIE) between the SAM and the perovskites is 0.42 eV for the Control group and 0.34 eV for the Target. The reduced ΔIE in the Target sample corresponds to a lower energetic barrier for hole extraction at the SAM/perovskite interface and improved energy-level alignment, facilitating hole transfer and helping suppress interfacial recombination (35).

To evaluate the influence of the hybrid SAM on device performance, we fabricated inverted PSCs with the configuration of ITO/NiO_x/SAMs/perovskite/fullerene (C₆₀)/SnO₂/silver (Ag). The cross-sectional SEM image in Fig. 4A shows a uniform and compact multilayer structure, confirming the successful deposition of each layer. To optimize the hybrid SAM composition, a series of Me-4PACz:TTA molar ratios was systematically investigated (fig. S22). Among these, the 3:1 ratio yields the best device performance and is therefore selected for subsequent studies. As shown in Fig. 4B, statistical analysis of the photovoltaic parameters indicates that devices based on the hybrid SAM exhibit notably higher open-circuit voltage (V_{OC}) and fill factor (FF) than the control group, exhibiting an average PCE of 22.1%, 1.8% higher than the Control (20.3%). The champion target device yields PCEs of 23.7 and 22.9% at reverse and forward scans, respectively (Fig. 4C). The detailed parameters are summarized in table S3, where the forward and reverse scans exhibit nearly identical V_{OC} (1.25 V) and J_{SC} (about 22.4 mA/cm²) values, along with FFs of

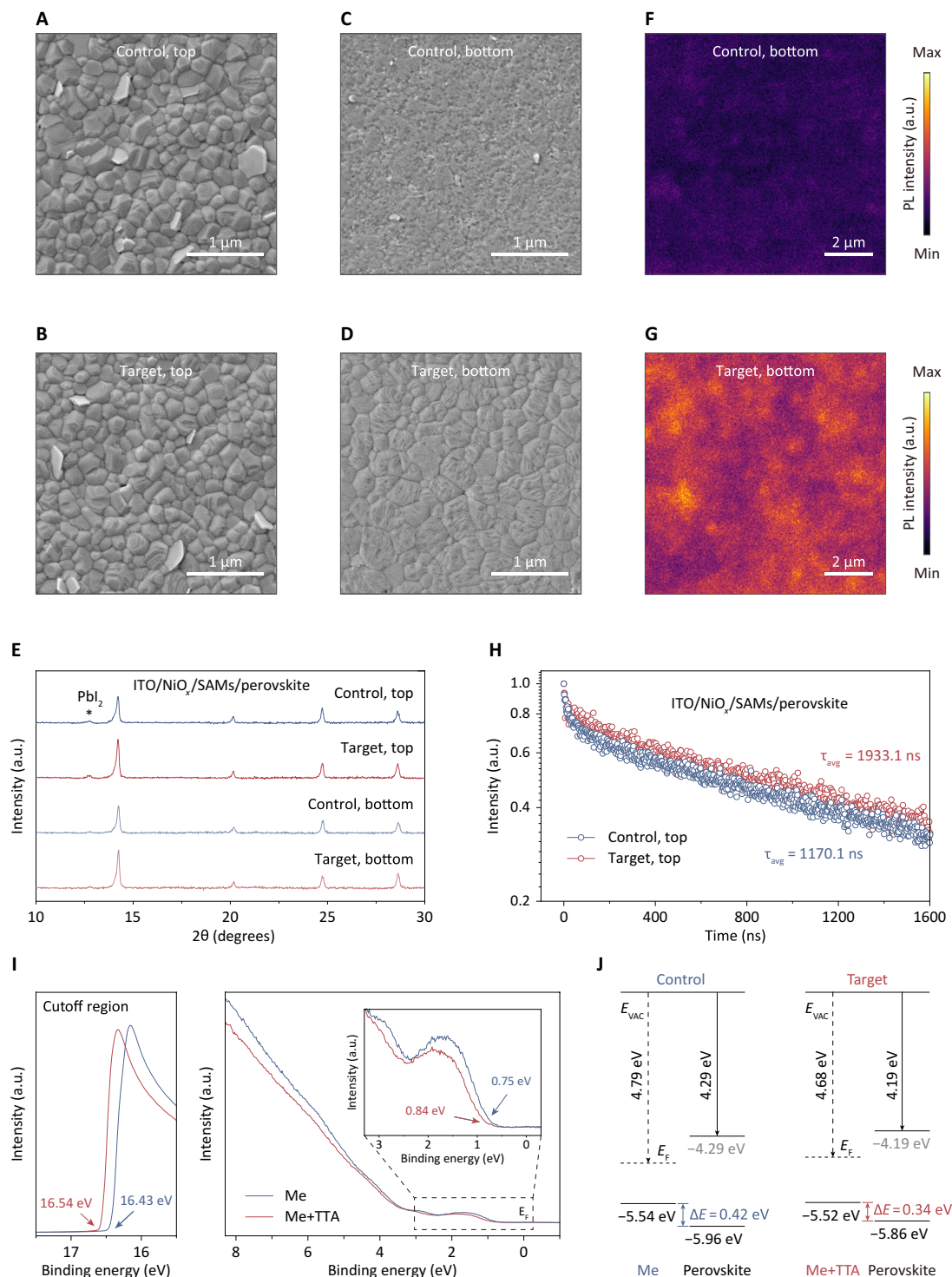


Fig. 3. Morphological and electronic properties of perovskite films with different SAMs. (A to D) Top and bottom SEM images of perovskite films on pure Me-4PACz and hybrid SAMs. (E) XRD patterns of perovskite films fabricated on pure Me-4PACz and hybrid SAMs. (F and G) PL mapping images of the control and target perovskite bottom surfaces. The perovskite films were deposited on pure Me-4PACz and hybrid SAMs and then peeled off for the PL characterization. (H) TRPL decay curves of perovskite films on pure Me-4PACz and hybrid SAMs. (I and J) UPS spectra of SAMs and perovskite layers used to evaluate interfacial energy-level alignment.

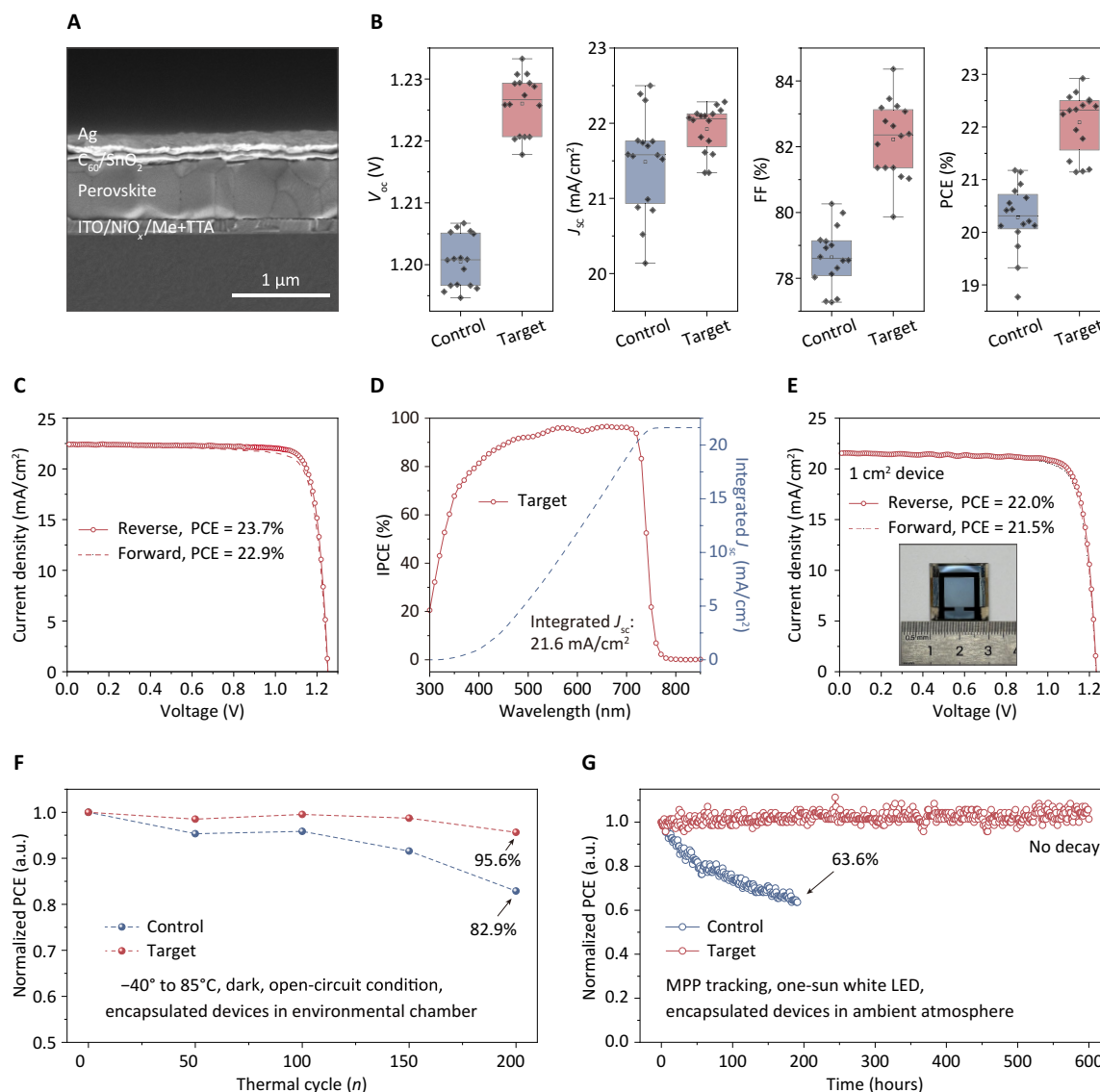


Fig. 4. Photovoltaic performance of single-junction devices. (A) Cross-sectional SEM image of the single-junction device. (B) Box plots comparing V_{OC} , J_{SC} , FF, and PCE values of the Control and Target devices. (C) J - V curves of the champion single-junction devices under reverse and forward scans. (D) IPCE spectrum and integrated J_{SC} of the Target device. (E) J - V curves of the champion 1-cm² large-area device under reverse and forward scans. (F) TC test from −40° to 85°C under dark conditions. (G) Steady-state power output measured under the MPPT condition. LED, light-emitting diode. SC, subscript.

81.6 and 84.3% (table S3). The corresponding incident photon-to-electron conversion efficiency (IPCE) spectrum and the integrated J_{SC} of 21.6 mA/cm² are in good agreement with the current density-voltage characteristic (J - V) results (Fig. 4D). Besides, for the large-area (1 cm²) devices (Fig. 4E), the reverse- and forward-scan efficiencies are 22.0 and 21.5%, respectively (detailed parameters are summarized in table S4), demonstrating the hybrid SAMs' consistent uniformity for scalability.

To test and compare device stability, a TC stability test (Fig. 4F) was conducted in an environmental chamber, where the devices were subjected to repeated temperature fluctuations between −40° and 85°C under dark conditions. After 200 cycles, the Control device (fig. S23) retained only 82.9% of its initial efficiency, whereas the Target device (fig. S24) maintained 95.6%, demonstrating its markedly

enhanced environmental durability. Under continuous maximum power point tracking (MPPT), the Target devices maintained 100% of their initial efficiency after 600 hours, whereas the Control devices decayed to 63.6% after 200 hours (Fig. 4G and figs. S25 and S26). Furthermore, the thermal stability tests at 85°C (fig. S27) indicate that the Target devices retained 90.8% of their initial efficiency after 1084 hours, whereas the Control devices decreased to 79.3% after 241 hours, clearly demonstrating the substantially enhanced thermal robustness of the Target devices.

To further evaluate the effect of the hybrid Me+TTA SAMs on device performance, perovskite/silicon tandem solar cells were fabricated. The schematic tandem device architecture is illustrated in Fig. 5A. The perovskite layer deposited on the textured silicon substrate with pyramid sizes of about 300 to 700 nm exhibits a uniform,

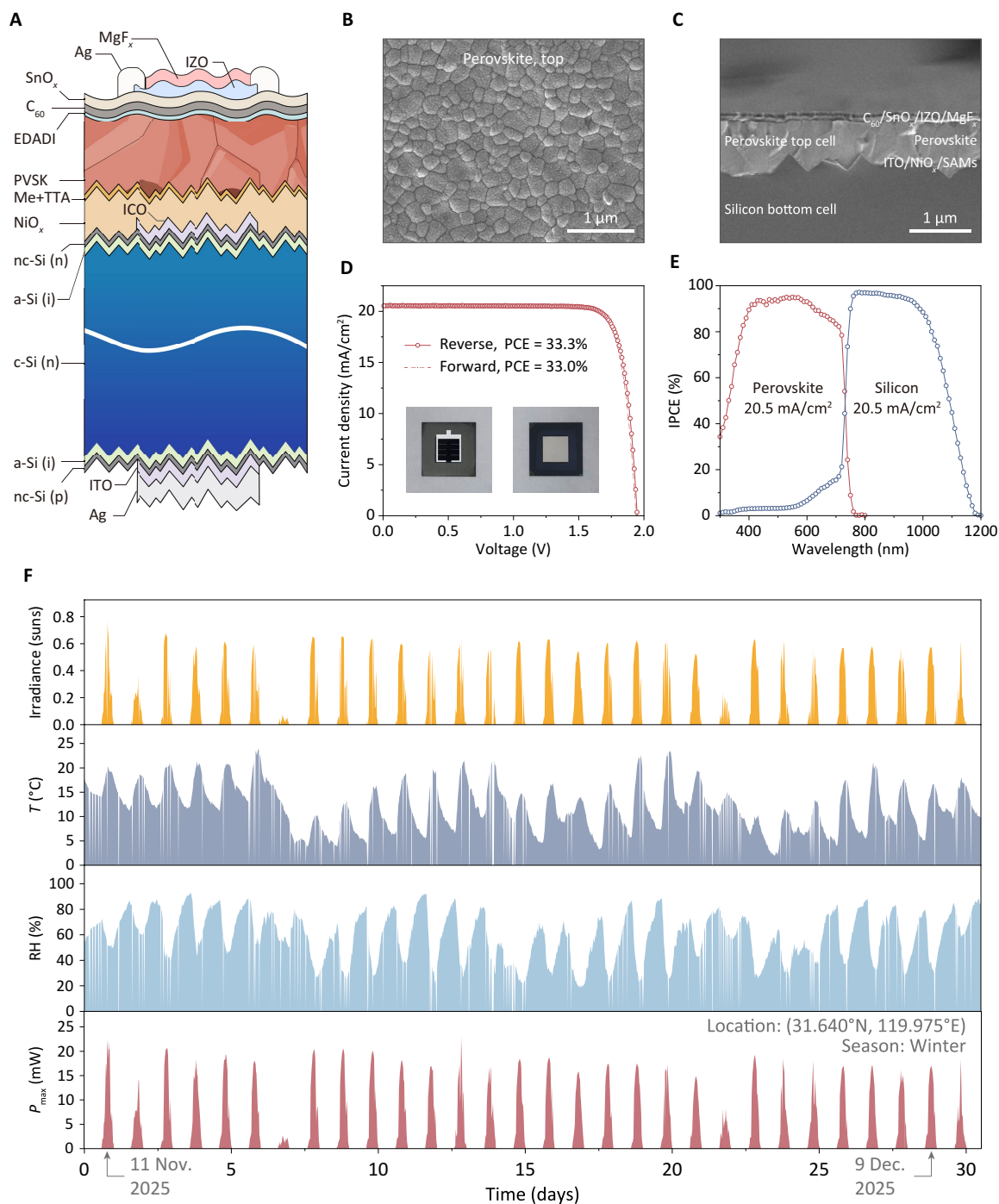


Fig. 5. Performance of perovskite/silicon tandem devices. (A) Schematic illustration of the perovskite/silicon tandem device architecture. (B) SEM image of the perovskite film surface on a silicon substrate. (C) Cross-sectional SEM image of the tandem device, showing the interface between the perovskite layer and the silicon substrates. (D) J - V curves of the champion tandem devices under reverse and forward scans. (E) IPCE spectra of the Target tandem device and the integrated J_{SC} . (F) Outdoor performance of the tandem device.

compact morphology (Fig. 5B), ensuring excellent interfacial contact between the top and bottom films. The cross-sectional SEM image (Fig. 5C) clearly shows well-defined interfaces and a continuous perovskite layer, confirming the successful integration of the hybrid SAM strategy in the tandem configuration. The champion tandem device incorporating the Me+TTA SAM achieved a PCE of 33.3%,

with a certified efficiency of 33.1%, as shown in Fig. 5D, fig.S28, and table S5. The IPCE spectrum and integrated J_{SC} are presented in Fig. 5E, showing excellent spectral response and confirming the accuracy of the measured J - V characteristics.

To evaluate the operational stability of the tandem device under realistic outdoor conditions, an outdoor durability test was carried

out using an encapsulated target device. During the measurement, diurnal variations in solar irradiance were continuously recorded, together with the ambient temperature and relative humidity (RH). Data acquisition was performed at 3-min intervals to ensure measurement reliability. As illustrated in Fig. 5F, the solar irradiance peaked at ~ 0.8 suns around noon on a clear day, whereas the ambient humidity fluctuated within the range of 30 to 90% RH. The instantaneous maximum power output ($P_{\max}$) closely followed the day-night variation of light intensity. Notably, after 30 days (about 720 hours) of continuous outdoor exposure with repeated day-night cycling, no discernible performance degradation was observed (figs. S29 and S30), confirming the device's excellent operational stability in outdoor environments. All devices used for long-term stability tests were encapsulated using a multilayer barrier structure, as shown in fig. S31 (details in the Materials and Methods section).

DISCUSSION

In summary, we have rationally designed a TTA molecule with tridentate carboxylic acid groups. The noncoplanar molecular configuration of TTA enables more ordered intermolecular packing between Me and TTA, thereby mitigating the intrinsic self-aggregation of Me-4PACz. This results in the formation of a more homogeneous, strongly bonded, and electronically favorable self-assembled molecular layer in WBG and tandem PSCs. The carboxylic acid and thiophene groups, assisted by a noncoplanar structure, allow free rotation and increase the probability of multisite binding to perovskite defects and adjacent layers, thereby improving crystallization and passivation. These combined effects result in notable improvement in photovoltaic performance and device stability. The optimized devices achieve higher V_{OC} , FF, and PCE values, together with superior operational stability under continuous illumination and elevated temperature. This work highlights a convenient yet versatile interfacial molecular engineering strategy that simultaneously regulates molecular packing, interfacial bonding, and perovskite crystallization. The Me-4PACz+TTA hybrid SAM approach provides insight into the design of stable and efficient WBG PSCs and offers a promising pathway toward high-performance perovskite/silicon tandem photovoltaics.

MATERIALS AND METHODS

Materials

Indium tin oxide (ITO) and fluorine-doped tin oxide (FTO) substrates, formamidinium iodide (FAI; 99.9%), and methylammonium bromide (MABr; 99.9%) were purchased from Advanced Election Technology Co. Ltd. Lead iodide (PbI_2 ; 99.99%), lead bromide ($PbBr_2$; 99.99%), and cesium iodide (CsI ; 99.9%) were purchased from Sigma-Aldrich. Rubidium thiocyanate ($RbSCN$; 99%) and 4-(3,6-dimethyl-9H-carbazol-9-yl)butane-1-phosphonic acid (Me-4PACz) were purchased from Suzhou Liwei New Material Co. Ltd. (China). C_{60} was purchased from Nano-C Co. Ltd. The solvents, including N,N' -dimethylformamide (DMF; 99.8%), dimethyl sulfoxide (DMSO; 99.9%), EtOH (99.9%), chlorobenzene (CB; 99.8%), and isopropanol (IPA; 99.8%), were purchased from Sigma-Aldrich. Ethanedi-amine dihydroiodide (EDADI; 99.5%) was purchased from Xi'an Yuri Solar Co. Ltd. Metal Ag, indium zinc oxide (IZO), indium cerium oxide (ICO), ITO, nickel oxide (NiO_x) targets were purchased from Zhongsheng Hengan (Beijing) New Material Technology. 5,5',5''-(nitritoltri-4,1-phenylene)tris[2-thiophenecarboxylic

acid] (TTA) was synthesized by our lab. All reagents and solvents were used without further purification.

WBG PSC fabrication

ITO substrates were sequentially ultrasonically cleaned in detergent, deionized water, and absolute EtOH for 30 min each, followed by nitrogen drying and ultraviolet ozone treatment for 30 min. Subsequently, an aqueous solution (10 mg mL^{-1}) of NiO_x was spin coated onto the substrates at 4000 rpm for 30 s, and the substrates were annealed at 150°C for 10 min in air. Alternatively, NiO_x films (about 15 nm in thickness) were sputtered from NiO_x targets, and the substrates were annealed at 150°C for 15 min in air. The substrates were then transferred into a nitrogen-filled glovebox for further processing.

Me-4PACz and TTA were used as SAM-based hole-transport materials. The Me-4PACz is dissolved in EtOH at 0.5 mg mL^{-1} , and the TTA is at 0.94 mg mL^{-1} (identical molar ratio). The control devices used pure Me-4PACz solution, whereas the target devices adopted a mixed Me-4PACz+TTA solution with a volume (molar) ratio of 3:1. The SAM layers were spin coated at 4000 rpm for 30 s, followed by thermal annealing at 120°C for 20 min. After that, the substrates were thoroughly rinsed with EtOH to remove weakly adsorbed or physically adsorbed molecules.

The perovskite precursor solution with a nominal composition of $Cs_{0.05}MA_{0.15}FA_{0.80}Pb(I_{0.75}Br_{0.25})_3$ (1.5 M) was prepared in a DMF:DMSO (volume ratio of 4:1) solvent mixture and stirred at 60°C for 2 hours. Subsequently, $RbSCN$ (2 mg mL^{-1}) was added as an additive. The precursor was spin coated at 4000 rpm for 40 s, and $110 \mu\text{L}$ of CB was dropped onto the film 10 s before the end of the spinning process. The films were then annealed at 100°C for 10 min. An IPA solution (0.25 mg mL^{-1}) containing EDADI was then spin coated onto the perovskite surface, followed by a postannealing step at 100°C for 5 min.

A 25-nm C_{60} layer and a 20-nm SnO_2 layer were sequentially deposited by atomic layer deposition (ALD), using tetrakis(dimethylamino) tin (TDMASn) [20 standard cubic centimeter per minute (SCCM), 0.1-s pulse, and 8.0-s purge] and H_2O (20 SCCM, 0.03-s pulse, and 8.0-s purge) as precursors. Last, a 200-nm Ag electrode was thermally evaporated on top of the device. To minimize optical losses, a 100-nm MgF_x antireflective coating was deposited on the front surface by thermal evaporation.

Perovskite/silicon tandem device fabrication

Silicon wafers with a thickness of about $150 \mu\text{m}$ were used. A double-sided random pyramid texture was obtained by immersing the wafers in KOH solution, yielding an average pyramid height of 500 to 700 nm on both sides. Surface passivation stacks were deposited using plasma-enhanced chemical vapor deposition (PECVD), including $\sim 6 \text{ nm}$ of $\alpha\text{-Si:H}$ at both sides, $\sim 12 \text{ nm}$ of n-doped nc-Si:H, and $\sim 12 \text{ nm}$ of p-doped nc-Si:H layers. The rear reflector consists of 110 nm of ITO by reactive plasma deposition. Silver (Ag) grid electrodes were then screen printed onto the rear surfaces, followed by annealing at 200°C to optimize electrical contact. On the front side, a 10-nm ICotunneling junction was sputtered through a mask with an opening area of 1.2 cm by 1.2 cm . Last, the wafers were laser scribed to a size of 2.5 cm by 2.5 cm . For the tandem fabrication, the bottom silicon substrates were subjected to ultraviolet ozone treatment for 7 min. Before perovskite deposition, a 15-nm layer of NiO_x was magnetron sputtered onto the front side of the silicon substrate. The as-prepared substrates were subjected to ultraviolet ozone

treatment for 30 min, followed by annealing at 200°C for 10 min. Subsequently, the SAM layers were spin coated onto the NiO_x surface under the same conditions as described for the single-junction perovskite devices.

The perovskite precursor solution with a nominal composition of Cs_{0.05}MA_{0.15}FA_{0.80}Pb(I_{0.75}Br_{0.25})₃ was prepared in a DMF:DMSO (volume ratio of 4:1) solvent mixture with a total concentration of 1.7 M and stirred at 60°C for 2 hours. Subsequently, RbSCN (2 mg ml⁻¹) was added as an additive. The precursor solution was spin coated at 4000 rpm for 40 s, during which 110 μl of CB was dropped onto the film 10 s before the end of the spinning process. The perovskite films were then annealed at 100°C for 10 min. A posttreatment solution containing EDADI (0.25 mg ml⁻¹ in IPA) was spin coated onto the perovskite surface, followed by postannealing at 100°C for 5 min.

After perovskite deposition, a 10-nm layer of the C₆₀ electron transport layer was thermally evaporated, followed by the deposition of a 10-nm SnO₂ layer by ALD. A 45-nm IZO transparent top electrode was subsequently deposited by magnetron sputtering. A Ag top electrode and a 100-nm layer of MgF_x antireflective coating were then sequentially deposited by thermal evaporation. Last, a Ag bottom electrode was thermally evaporated onto the rear side of the silicon substrate to complete the perovskite/silicon tandem devices.

Characterization of the SAM and perovskite layers

The chemical composition and bonding states of the SAM layers were analyzed by XPS using a Thermo Fisher Scientific Al K-Alpha system, focusing on the S 2p and N 1s regions to confirm TTA incorporation. Depth profiles of the ITO/NiO_x/SAM/perovskite stack were obtained using TOF-SIMS to examine molecular distribution and interface integrity. Cross-sectional TEM was used to assess the morphology and uniformity of the SAM layers. C-AFM was used to map local current distribution across the SAM films, whereas KPFM measured surface potential before and after thermal aging at 110°C for 8 hours. Interfacial adhesion between the SAM and perovskite layers was quantified using a tensile testing setup. The interfacial tensile strength of the SAM/perovskite interface was evaluated using a pull-off test on perovskite films. The perovskite films were covered using epoxy resin. Metal modules were attached to both sides of the samples using glue, with the covered area intentionally smaller than the perovskite film area to ensure that the applied load was localized to the target interface. The samples were then subjected to vertical tensile loading using a load-displacement control system (MTS CMT6104) equipped with a 10-kN load cell at a constant loading rate of 0.2 mm/min. The recorded force-displacement curves were used to determine the maximum pull-off force. The interfacial tensile strength was calculated on the basis of the maximum force normalized by the effective contact area. Only samples exhibiting interfacial fracture at the SAM/perovskite interface were included in the analysis. The morphology of perovskite films on different SAMs was characterized by SEM using a TESCAN microscope, and crystallographic properties were assessed by XRD using a PANalytical B.V. X'Pert Pro MPD x-ray diffractometer with Cu Kα radiation. PL mapping and TRPL measurements were performed using an Edinburgh FLS 1000 spectrometer, with 425-nm laser excitation (~5 nJ/cm²) incident from the perovskite side. SCLC measurements were carried out to evaluate the trap density in the perovskite films. Hole-only devices with a structure of ITO/NiO_x/SAMs/perovskite/Au were fabricated for this purpose. The current-voltage (*I*-*V*) characteristics were measured under dark

conditions. The *V*_{TFL} was extracted from the kink point in the *I*-*V* curves, corresponding to the transition from the ohmic region to the trap-filled-limited regime. The *N*_{trap} was calculated using the standard SCLC model according to the following equation

$$N_{\text{trap}} = \frac{2\epsilon_0\epsilon_r V_{\text{TFL}}}{eL^2} \quad (1)$$

where ϵ_0 is the vacuum permittivity, ϵ_r is the relative dielectric constant of the perovskite, e is the elementary charge, and L is the thickness of the perovskite layer, which is ~600 nm. All samples share the same perovskite composition and thickness, allowing for a direct comparison of trap densities. UPS with a He I (21.22 eV) source was used to evaluate the interfacial energy-level alignment. The femtosecond TA measurements are performed using a fiber laser (1030 nm, 100-kHz repetition rate, 9.5 μJ/pulse; YF-FL-10-100-IR, Yacto-Technology, China) as the laser source and a femto-TA100 spectrometer (Time-Tech Spectra, China). The AFM-IR mapping was measured by a Bruker nanoIR3 with contact mode. For the characteristic vibrational signals, Me-4PACz is 1105 cm⁻¹ and TTA is 1665 cm⁻¹.

Characterization of PSCs

Photovoltaic parameters (*V*_{OC}, *J*_{SC}, FF, and PCE) of the single-junction and tandem devices were measured under AM 1.5G (air mass 1.5 global solar spectrum) illumination (100 mW/cm²) using a Keithley 2400 source meter and a calibrated xenon lamp solar simulator (EnliTech, SRC2020), with an active area defined by an aperture mask of 0.0625 cm² for the single-junction cell and 0.999 cm² for the perovskite/silicon tandem cells. Reverse and forward scans were conducted. IPCE spectra were collected using a QE-R system (EnliTech) and integrated to obtain the *J*_{SC}. TC stability was tested by exposing devices to repeated -40° to 85°C cycles in the dark. Continuous MPPT under 1-sun illumination (light-emitting diode) was performed to evaluate steady-state power output. Outdoor operational stability was assessed by monitoring encapsulated devices under natural sunlight for 30 days.

Encapsulation of devices

For long-term stability tests, the devices were encapsulated using a multilayer barrier structure. A 2-mm-thick glass cover (5 cm by 5 cm) was used as the top substrate. Polyolefin elastomer (POE) films were used as the encapsulant, and polyisobutylene (PIB) was used as the edge sealant. Copper ribbons were first attached to the device electrodes using silver paste, followed by a preheating process at 90°C for 3 min to improve interfacial adhesion. Subsequently, the encapsulation stack (glass/POE/device/POE/PIB) was assembled and transferred to a laminator. The lamination was carried out under sequential pressures of 30 to 50 kPa at 105°C, with vacuum and lamination durations of 360 and 400 s, respectively. After lamination, the encapsulated devices were used for stability measurements.

DFT calculations

We constructed the (001) surface of the perovskite and the (100) surface of the NiO, based on a 3 by 3 by 2 FAPbI_{0.75}Br_{0.25} slab and a 6 by 6 by 2 NiO slab. The slabs are $a = b = 25.6295 \text{ \AA}$, $c = 20.5065 \text{ \AA}$, and $\alpha = \beta = \gamma = 90^\circ$ for FAPbI_{0.75}Br_{0.25} and $a = b = c = 20.6325 \text{ \AA}$ and $\alpha = \beta = \gamma = 90^\circ$ for NiO. The SAM molecules were initially placed in the *z* direction away from the surface with a distance of 2.5 Å. We used a vacuum of 15 Å to separate neighboring surfaces in

the z direction. The Vienna ab initio simulation package (VASP) was used for ab initio DFT calculations. The projector-augmented-wave (PAW) potentials with a kinetic energy cutoff of 520 eV and the exchange-correlation form in the Perdew-Burke-Eruzerhof (PBE) generalized gradient approximation (GGA) were used. The energy and force convergence criteria were set to 10^{-5} eV and -0.05 eV Å $^{-1}$, respectively. The Brillouin zone was sampled with a Γ -centered Monkhorst-Pack 1 by 1 by 1 k -point grid. In addition, we used Grimme's DFT-D3 method for the van der Waals correction. The ionic optimizations fixed the bottom two atom layers to represent the bulk structures. The binding energies (E_b) of different molecules with the perovskite surface were calculated as

$$E_b = E_{\text{mol/pvsk}} - E_{\text{pvsk}} - E_{\text{mol}} \quad (2)$$

where $E_{\text{mol/pvsk}}$, E_{pvsk} , and E_{mol} are the total energies of the adsorption system, the perovskite, and the SAM molecule, respectively. The NiO system was the same as the previous calculation process.

The dimer model of SAM molecules was constructed by placing two molecules at an initial interlayer distance of 2.5 Å and in a typical parallel configuration based on π - π stacking. Geometric optimization was carried out with convergence criteria of 10^{-5} eV for energy and -0.05 eV/Å for force. Vacuum layers of 15 Å were used to separate neighboring surfaces in the x , y , and z directions respectively. The binding energies (E_b) were calculated as

$$E_b = \frac{E_{\text{dimer}} - 2 \cdot E_{\text{monomer}}}{2} \quad (3)$$

where E_{dimer} and E_{monomer} are the electronic energy of the dimer cluster and the single molecule, respectively.

Supplementary Materials

This PDF file includes:

Figs. S1 to S31

Tables S1 to S5

REFERENCES

- X. Y. Chin, D. Turkay, J. A. Steele, S. Tabean, S. Eswara, M. Mensi, P. Fiala, C. M. Wolff, A. Paracchino, K. Artuk, D. Jacobs, Q. Guesnay, F. Sahli, G. Andreatta, M. Boccard, Q. Jeangros, C. Ballif, Interface passivation for 31.25%-efficient perovskite/silicon tandem solar cells. *Science* **381**, 59–63 (2023).
- Y. Chen, N. Yang, G. Zheng, F. Pei, W. Zhou, Y. Zhang, L. Li, Z. Huang, G. Liu, R. Yin, H. Zhou, C. Zhu, T. Song, C. Hu, D. Zheng, Y. Bai, Y. Duan, Y. Ye, Y. Wu, Q. Chen, Nuclei engineering for even halide distribution in stable perovskite/silicon tandem solar cells. *Science* **385**, 554–560 (2024).
- Q. Jiang, J. Tong, R. A. Scheidt, X. Wang, A. E. Louks, Y. Xian, R. Tirawat, A. F. Palmstrom, M. P. Hautzinger, S. P. Harvey, S. Johnston, L. T. Schelhas, B. W. Larson, E. L. Warren, M. C. Beard, J. J. Berry, Y. Yan, K. Zhu, Compositional texture engineering for highly stable wide-bandgap perovskite solar cells. *Science* **378**, 1295–1300 (2022).
- Z. Fang, B. Deng, Y. Jin, L. Yang, L. Chen, Y. Zhong, H. Feng, Y. Yin, K. Liu, Y. Li, J. Zhang, J. Huang, Q. Zeng, H. Wang, X. Yang, J. Yang, C. Tian, L. Xie, Z. Wei, X. Xu, Surface reconstruction of wide-bandgap perovskites enables efficient perovskite/silicon tandem solar cells. *Nat. Commun.* **15**, 10554 (2024).
- Z. Liu, R. Lin, M. Wei, M. Yin, P. Wu, M. Li, L. Li, Y. Wang, G. Chen, V. Carnevali, L. Agosta, V. Slama, N. Lempesis, Z. Wang, M. Wang, Y. Deng, H. Luo, H. Gao, U. Rothlisberger, S. M. Zakeeruddin, X. Luo, Y. Liu, M. Grätzel, H. Tan, All-perovskite tandem solar cells achieving >29% efficiency with improved (100) orientation in wide-bandgap perovskites. *Nat. Mater.* **24**, 252–259 (2025).
- H. Chen, C. Liu, J. Xu, A. Maxwell, W. Zhou, Y. Yang, Q. Zhou, A. S. R. Bati, H. Wan, Z. Wang, L. Zeng, J. Wang, P. Serles, Y. Liu, S. Teale, Y. Liu, M. I. Saidaminov, M. Li, N. Rolston, S. Hoogland, T. Filleter, M. G. Kanatzidis, B. Chen, Z. Ning, E. H. Sargent, Improved charge extraction in inverted perovskite solar cells with dual-site-binding ligands. *Science* **384**, 189–193 (2024).
- H. Tang, Z. Shen, Y. Shen, G. Yan, Y. Wang, Q. Han, L. Han, Reinforcing self-assembly of hole transport molecules for stable inverted perovskite solar cells. *Science* **383**, 1236–1240 (2024).
- D. Li, Q. Lian, T. Du, R. Ma, H. Liu, Q. Liang, Y. Han, G. Mi, O. Peng, G. Zhang, W. Peng, B. Xu, X. Lu, K. Liu, J. Yin, Z. Ren, G. Li, C. Cheng, Co-adsorbed self-assembled monolayer enables high-performance perovskite and organic solar cells. *Nat. Commun.* **15**, 7605 (2024).
- F. Yuan, T. Xue, M. Du, H. Huang, R. Zeng, L. Li, C. Wang, Z. Song, Q. Guo, X. Hu, E. Zhou, Post-assembled dipole benzoic acids modified Me-4PCz for efficient and stable inverted perovskite solar cells. *Adv. Funct. Mater.* **35**, 2425145 (2025).
- S. M. Park, M. Wei, N. Lempesis, W. Yu, T. Hossain, L. Agosta, V. Carnevali, H. R. Atapattu, P. Serles, F. T. Eickemeyer, H. Shin, M. Vafaie, D. Choi, K. Darabi, E. D. Jung, Y. Yang, D. B. Kim, S. M. Zakeeruddin, B. Chen, A. Amassian, T. Filleter, M. G. Kanatzidis, K. R. Graham, L. Xiao, U. Rothlisberger, M. Grätzel, E. H. Sargent, Low-loss contacts on textured substrates for inverted perovskite solar cells. *Nature* **624**, 289–294 (2023).
- K. Hossain, A. Kulkarni, B. Bothra, B. Klingebiel, T. Kirchartz, M. Saliba, D. Kabra, Resolving the hydrophobicity of the Me-4PCz hole transport layer for inverted perovskite solar cells with efficiency >20%. *ACS Energy Lett.* **8**, 3860–3867 (2023).
- H. Zhou, K. Cai, S. Yu, Z. Wang, Z. Xiong, Z. Chu, X. Chu, Q. Jiang, J. You, Efficient and stable perovskite mini-module via high-quality homogeneous perovskite crystallization and improved interconnect. *Nat. Commun.* **15**, 6679 (2024).
- P. Caprioglio, J. A. Smith, R. D. J. Oliver, A. Dasgupta, S. Choudhary, M. D. Farrar, A. J. Ramadan, Y.-H. Lin, M. G. Christoforo, J. M. Ball, J. Diekmann, J. Thiesbrummel, K.-A. Zaininger, X. Shen, M. B. Johnston, D. Neher, M. Stollerfoht, H. J. Snaith, Open-circuit and short-circuit loss management in wide-gap perovskite p-i-n solar cells. *Nat. Commun.* **14**, 932 (2023).
- C. Li, Y. Chen, Z. Zhang, C. Liu, F. Guo, W. Ahmad, P. Gao, Pros and cons of hole-selective self-assembled monolayers in inverted PSCs: Extensive case studies and data analysis. *Energy Environ. Sci.* **17**, 6157–6203 (2024).
- Y. Yuan, H. Li, H. Luo, Y. Zhang, X. Li, T. Jiang, Y. Yang, L. Liu, B. Fan, X. Hao, A comprehensive review of self-assembled monolayers as hole-transport layers in inverted perovskite solar cells. *Energies* **18**, 2577 (2025).
- J. Wang, S. Hu, H. Zhu, S. Liu, Z. Zhang, R. Chen, J. Wang, C. Shi, J. Zhang, W. Liu, X. Lei, B. Liu, Y. Pan, F. Ren, H. Raza, Q. Zhou, S. Li, L. Qiu, G. Zheng, X. Qin, Z. Zhao, S. Yang, N. Li, J. Li, A. Wakamiya, Z. Liu, H. J. Snaith, W. Chen, Mercapto-functionalized scaffold improves perovskite buried interfaces for tandem photovoltaics. *Nat. Commun.* **16**, 4917 (2025).
- G. Qu, S. Cai, Y. Qiao, D. Wang, S. Gong, D. Khan, Y. Wang, K. Jiang, Q. Chen, L. Zhang, Y.-G. Wang, X. Chen, A. K. Y. Jen, Z.-X. Xu, Conjugated linker-boosted self-assembled monolayer molecule for inverted perovskite solar cells. *Joule* **8**, 2123–2134 (2024).
- Y. Luo, Y. Tian, K. Zhao, W. Mao, C. Liu, J. Shen, Z. Cheng, C. Değer, X. Miao, Z. Zhang, X. Sun, L. Yao, X. Zhang, P. Shi, D. Jin, J. Deng, M. Tian, I. Yavuz, N. Dong, R. Liu, R. Wang, D. Yang, J. Xue, Inductive effects in molecular contacts enable wide-bandgap perovskite cells for efficient perovskite/TOPCon tandems. *Nat. Commun.* **16**, 4516 (2025).
- S. Liu, J. Li, W. Xiao, R. Chen, Z. Sun, Y. Zhang, X. Lei, S. Hu, M. Kober-Czerny, J. Wang, F. Ren, Q. Zhou, H. Raza, Y. Gao, Y. Ji, S. Li, H. Li, L. Qiu, W. Huang, Y. Zhao, B. Xu, Z. Liu, H. J. Snaith, N.-G. Park, W. Chen, Buried interface molecular hybrid for inverted perovskite solar cells. *Nature* **632**, 536–542 (2024).
- S. Li, Y. Jiang, J. Xu, D. Wang, Z. Ding, T. Zhu, B. Chen, Y. Yang, M. Wei, R. Guo, Y. Hou, Y. Chen, C. Sun, K. Wei, S. M. H. Qaid, H. Lu, H. Tan, D. Di, J. Chen, M. Grätzel, E. H. Sargent, M. Yuan, High-efficiency and thermally stable FACsPbI $_3$ perovskite photovoltaics. *Nature* **635**, 82–88 (2024).
- J. J. Plata, E. J. Blancas, A. M. Márquez, V. Posligua, J. Fdez Sanz, R. Grau-Crespo, Harnessing the unusually strong improvement of thermoelectric performance of AgInTe $_2$ with nanostructuring. *J. Mater. Chem. A* **11**, 16734–16742 (2023).
- Q. Zhang, Q. Zhao, H. Wang, Y. Yao, L. Li, Y. Wei, R. Xu, C. Zhang, E. O. Shalenov, Y. Tu, K. Wang, M. Xiao, Tuning isomerism effect in organic bulk additives enables efficient and stable perovskite solar cells. *Nanomicro Lett.* **17**, 107 (2025).
- S. Zhang, X. Wang, Y. Wu, X. Li, T. Hou, D. Li, W. Chen, J. Li, R. Lv, Y. Zhang, Z. Xiao, T. Chen, S. Chen, Z. Fang, S. Liu, S. Yang, Self-assembled π -conjugated hole-selective molecules for UV-resistant high-efficiency perovskite solar cells. *Angew. Chem. Int. Ed. Engl.* **64**, e202508782 (2025).
- S. Ameen, D. Lee, A. B. Faheem, J. G. Son, Y. Lee, H. Yoo, S. Park, Y. S. Shin, J. Lee, J. Seo, H. Jang, J. Roe, J. W. Song, S. Cho, Y. Park, K.-K. Lee, J. Y. Kim, D. S. Kim, B. Kim, 3,6-Bis(methylthio)-9H-carbazole based self-assembled monolayer for highly efficient and stable inverted perovskite solar cells. *Angew. Chem. Int. Ed. Engl.* **64**, e202423206 (2025).
- M. B. Faheem, Y. Zhang, M. B. Saud, H. Li, P. I. Kaswekar, Q. Qiao, Nanoscale spatial and time-resolved mapping in perovskite and organic solar cells: A multimodal technique to visualize the photoinduced charge dynamics. *J. Mater. Chem. A* **12**, 16329–16342 (2024).
- B. Zhang, J. Luo, H. Yin, Q. Li, S. Sun, N. Zhang, N. Gan, M. Azam, T. W. Park, Z. Wan, C. Jia, M. Wei, S. M. Park, A cross-linked molecular contact for stable operation of perovskite/silicon tandem solar cells. *Science* **390**, 837–842 (2025).

27. S. Ning, Z. Li, L. Sun, X. Dong, X. Xu, R. Mao, M. Wang, Z. Li, P. Zhu, J. Zhu, Reinforcing the buried interface enables efficient and stable wide-bandgap perovskite solar cells for perovskite/silicon tandems. *Adv. Opt. Mater.* **13**, e02504 (2025).
28. Y. Zhao, Y. Liu, Z. Ren, Y. Li, Y. Zhang, F.-C. Kong, T. Liu, X. Shi, Y. Dou, L. Wang, F. Wang, X. Guo, Y. Cao, W. Wang, P. C. Y. Chow, S. Chen, Enhanced interface adhesion with a polymeric hole transporter enabling high-performance air-processed perovskite solar cells. *Energy Environ. Sci.* **18**, 1366–1374 (2025).
29. D. Song, S. Ramakrishnan, Y. Zhang, Q. Yu, Mixed self-assembled monolayers for high-photovoltage tin perovskite solar cells. *ACS Energy Lett.* **9**, 1466–1472 (2024).
30. C. Shi, J. Wang, X. Lei, Q. Zhou, W. Wang, Z. Yang, S. Liu, J. Zhang, H. Zhu, R. Chen, Y. Pan, Z. Tan, W. Liu, Z. Zhao, Z. Cai, X. Qin, Z. Zhao, J. Li, Z. Liu, W. Chen, Modulating competitive adsorption of hybrid self-assembled molecules for efficient wide-bandgap perovskite solar cells and tandems. *Nat. Commun.* **16**, 3029 (2025).
31. Y. Zhang, X. Dong, L. Wang, Y. Niu, P. Wang, M. Wang, Y. Yan, Y. Dou, X. Shi, Q. Zhang, J. Wang, S. Chen, P. Zhu, J. Zhu, Multifunctional phenothiazine-based self-assembled monolayer as a hole-selective contact for efficient wide-band-gap perovskite solar cells. *Nano Lett.* **25**, 11257–11265 (2025).
32. J. Thiesbrummel, S. Shah, E. Gutierrez-Partida, F. Zu, F. Peña-Camargo, S. Zeiske, J. Diekmann, F. Ye, K. P. Peters, K. O. Brinkmann, P. Caprioglio, A. Dasgupta, S. Seo, F. A. Adeleye, J. Warby, Q. Jeangros, F. Lang, S. Zhang, S. Albrecht, T. Riedl, A. Armin, D. Neher, N. Koch, Y. Wu, V. M. Le Corre, H. Snaith, M. Stollerfoht, Ion-induced field screening as a dominant factor in perovskite solar cell operational stability. *Nat. Energy* **9**, 664–676 (2024).
33. Y. Niu, S. Lin, X. Dong, M. Wang, Y. Zhang, S. Ning, Z. Li, J. Wang, J. Yin, S. Chen, P. Zhu, J. Zhu, Homogenizing strain via reinforced lattice interaction enables efficient and stable 4T perovskite/silicon tandem solar cells. *Joule* **9**, 102142 (2025).
34. J.-A. Li, H. Ma, X. Ding, S. Li, B. Han, W. Chen, S. Huang, S. Chen, Y. Kuang, Z. Liu, C. Yan, Synergistic passivation and stable carrier transport enable efficient blade-coated perovskite solar cells fabricated in ambient air. *J. Mater. Chem. A* **13**, 30140–30150 (2025).
35. J. Yang, G. Qu, Y. Qiao, S. Cai, J. Hu, S. Geng, Y. Li, Y. Jin, N. Shen, S. Chen, A. K. Jen, Z. X. Xu, Flexibility meets rigidity: A self-assembled monolayer materials strategy for perovskite solar cells. *Nat. Commun.* **16**, 6968 (2025).

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